

EDUCATION

SunYat-sen University

Bachelor of Science in Software Engineering; **Expected GPA: 3.8 / 4.00**

Zhuhai, China

Sep. 2023 - Present

Core Courses: Computer Programming, Machine Learning and Data Mining, Natural Language Processing, Operating Systems, Digital Circuits, Software Analysis and Design, Digital Image Processing, Algorithm Design and Analysis, Database Systems

PUBLICATIONS

Publication: Xu, D., Liu, M., **Wang, Xiwen**, Zhong, J., & Zheng, Z. "RepoDoc: A Knowledge Graph-Based Framework to Automatic Documentation Generation and Incremental Updates." *arXiv:2604.26523*, 2026.

Zheng, D., Wang, Y., **Wang, Xiwen**, Duan, K., Zhang, H., Liu, X., Ma, Y., & Zheng, Z. "From Static to Dynamic: Benchmarking Real-World Code Review with MCR-Bench." *arXiv:2608.27442*, 2026.

RESEARCH EXPERIENCE

RepoDoc: A Knowledge Graph-Based Framework to Automatic Documentation Generation and Incremental Updates

Research Assistant

Sep. 2025 – Apr. 2026

- Supervised by Prof. Mingwei Liu to develop an LLM-based agent, *RepoDoc*, and contributed to the development of a framework for large code repositories, which reduced token consumption by 85% and accelerated documentation generation by 3×
- Assisted to design the repository knowledge graph framework by organizing code entities and dependencies into connected modules, using AST parsing and LLM-based semantic enrichment, to retrieve relevant code context and update affected documentation
- Designed and refined the Vue.js documentation interface by integrating Mermaid.js-based workflow, architecture, and call-graph visualizations, enabling developers to intuitively understand repository structure and execution flows and reducing the onboarding barrier for newcomers
- Contributed to RepoDocBench by designing its Text Quality Score evaluation protocol with LLM-as-a-Judge, developing prompts and few-shot scoring guidelines for assessing documentation quality, and conducting follow-up benchmark exploration through the collection and evaluation of repository-level documentation

From Static to Dynamic: Benchmarking Real-World Code Review with MCR-Bench

Research Assistant

Oct. 2025 - Jan. 2026

- Supervised by Prof. Yanlin Wang to develop MCR-Bench, the first defect state-aware benchmark for realistic multi-round code review, comprising 2,269 real-world review tasks across five programming languages
- Participated in a comparative survey of 8 prior automated code-review benchmarks, identifying limitations of single-round static evaluation and requirements for cross-round interaction modeling and defect-state tracking
- Contributed to developing Python-based GitHub data-mining pipelines for the Java and Python subsets, processing 14,029 merged PRs (10,546 Python; 3,483 Java) and stratifying histories by 2–9 review rounds, with up to 500 instances sampled per round per language
- Co-designed a structured JSON-based Defect Card schema encoding defect description, file/line location, lifecycle state, taxonomy and severity, enabling state-aware modeling of defect evolution across multiple review rounds rather than isolated single-round review

RELEVANT PROJECTS

Skill Learning for SWE Agents

Zhuhai, China

Project Leader

Sep. 2026 – Present

- Build up a skill-learning framework to extract, organize, and reuse transferable skills from the multi-step execution trajectories of real-world software engineering agents, which improved skill selection across different stages of coding tasks
- Designed a stage-aware skill-bank framework that partitions skills into early-stage context understanding and task localization, mid-stage code modification and debugging, and late-stage testing and validation, enabling phase-conditioned skill generation, storage, and retrieval rather than stage-agnostic top-k retrieval.

PianoPal, NUS School of Computing Summer Workshop

Singapore

Project Leader

Jun. 2026 – Aug. 2026

- Led a four-member team to develop PianoPal, an AI- augmented piano practice companion that provides real-time feedback on posture, pitch, and rhythm, and earned Second Place among 11 teams in the AIoT course within the 458-participant NUS Summer Workshop
- Designed a front-end dashboard that interacted with a Raspberry Pi via RESTful APIs, with a Python edge backend running entirely on the Pi to process microphone and BLE IMU sensor inputs and control speaker and LED-strip outputs
- Developed a posture-monitoring pipeline using wrist and finger-mounted IMU sensors to collect motion data and a Random Forest classifier to identify improper playing behaviors, including wrist collapse, excessive wrist arching, etc.
- Implemented an audio-based performance evaluation module using microphone for WAV audio capture and Librosa-based onset detection and Constant-Q Transform (CQT) pitch analysis to extract played notes and onset times, and fine-tuned

Dynamic Time Warping (DTW) algorithm to detect pitch errors, rhythm deviations, missed notes, and extra notes

AI-assisted design of HVAC systems and HMI engineering drawings, *Electrical & Electronic Engineering Innovation Competition, Schneider Electric*

Guangzhou, China

Team Leader & Developer

Apr. 2025 – Aug. 2025

- Co-developed an AI system to automatically generate Human-Machine Interface (HMI) diagrams from industrial-process images and user requirements, which earned National First Prize in the competition's AI Track
- Applied the Segment Anything Model (SAM) with click-guided prompting and adapted inference APIs to segment industrial components from uploaded images, converting visual inputs into reusable component assets for subsequent HMI generation.
- Integrated an LLM-based planning module using API-based structured prompting with user requirements, component metadata, aesthetic guidelines, and engineering constraints to determine HMI component placement and interconnections, reducing manual interface-planning time by around 50%.
- Designed a structured JSON schema to archive component types, coordinates, dimensions, connections, and visual properties, enabling LLM-generated layout plans to be converted into executable VXML engineering files and rendered HMI diagrams.

SKILLS

Computational Skills: Java, React, Spring, thymeleaf, HTML, CSS, JavaScript, TypeScript, C#, ASP .NET, Swing, Python, SQL, DrRacket, Redux, Latex

Software: NPM, Spring Boot, Git, IntelliJ, Pycharm, Eclipse, Visual Studio, Emacs, VirtualBox